

Math 421: Homework 8

Solutions

Due: Nov 7, 2025

Problem 1: Prove, directly from the definition, that (s_n) where $s_n = \frac{1}{n}$ is a Cauchy sequence.

Proof. Let $\epsilon > 0$. Set $N = 2/\epsilon$. Assume $n, m > N$, then

$$|s_n - s_m| = \left| \frac{1}{n} - \frac{1}{m} \right| \leq \left| \frac{1}{n} \right| + \left| -\frac{1}{m} \right| = \frac{1}{n} + \frac{1}{m} < \frac{1}{N} + \frac{1}{N} = \frac{2}{N} = \frac{2}{2/\epsilon} = \epsilon.$$

Thus, since $\epsilon > 0$ was arbitrary (s_n) is a Cauchy sequence. $\square$

Problem 2: Let $x_n = \sqrt{n}$.

(a) *Proof.*

$$\begin{aligned} \lim |x_{n+1} - x_n| &= \lim |\sqrt{n+1} - \sqrt{n}| \\ &= \lim \left| (\sqrt{n+1} - \sqrt{n}) \cdot \frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1} + \sqrt{n}} \right| \\ &= \lim \left| \frac{(n+1) - n}{\sqrt{n+1} + \sqrt{n}} \right| \\ &= \lim \left| \frac{1}{\sqrt{n+1} + \sqrt{n}} \right| = \lim \frac{1}{\sqrt{n+1} + \sqrt{n}} = 0 \end{aligned}$$

$\square$

(b) Show, directly from the definition, that (x_n) is not a Cauchy sequence.

Proof. To prove that (x_n) is not a Cauchy sequence, we must prove that there exists an ϵ such that for every $N \in \mathbb{N}$ there exists $n, m \geq N$ such that $|x_n - x_m| \geq \epsilon$. Let $\epsilon = 1$ and set $n = (N+2)^2 > N$ and $m = (N+1)^2 > N$. Then

$$|x_n - x_m| = \left| \sqrt{(N+2)^2} - \sqrt{(N+1)^2} \right| = |(N+2) - (N+1)| = 1 \geq \epsilon.$$

$\square$

Problem 3: (Ross 10.6 a)

Theorem. Let (s_n) be a sequence such that $|s_{n+1} - s_n| < 2^{-n}$ for all $n \in \mathbb{N}$. (s_n) is a Cauchy sequence and hence a convergent sequence.

Proof. Let (s_n) be a sequence such that $|s_{n+1} - s_n| < 2^{-n}$ for all $n \in \mathbb{N}$. Let $\epsilon > 0$ and set $N = \log_2 2/\epsilon$. Then for all $n, m > N$, assume, without loss of generality $m > n$, we have

$$\begin{aligned} |s_m - s_n| &= |s_m - s_{m-1} + s_{m-1} - s_{m-2} + \cdots + s_{n+1} - s_n| \\ &\leq |s_m - s_{m-1}| + |s_{m-1} - s_{m-2}| + \cdots + |s_{n+1} - s_n| \end{aligned}$$

by the triangle inequality. By the hypothesis on consecutive terms in the sequence, we have

$$|s_m - s_{m-1}| + |s_{m-1} - s_{m-2}| + \cdots + |s_{n+1} - s_n| < 2^{-(m-1)} + 2^{-(m-2)} + \cdots + 2^{-n}.$$

Note that this is a geometric series

$$\sum_{i=n}^{m-1} 2^{-i} \leq \lim_{k \rightarrow \infty} 2^{-n} \sum_{i=1}^k 2^{-i} = \frac{2^{-n}}{1 - 2^{-1}} = \frac{2}{2^n} < \frac{2}{2^N} = \frac{2}{2^{\log_2 2/\epsilon}} = \frac{2}{2/\epsilon} = \epsilon$$

Thus,

$$|s_m - s_n| < \epsilon$$

and (s_n) is a Cauchy sequence and hence a convergent sequence. $\square$

Problem 4: (Ross 14.6)

Theorem. *Prove that if $\sum |a_n|$ converges and (b_n) is a bounded sequence, then $\sum a_n b_n$ converges.*

Proof. Let (b_n) be a bounded sequence. Then there exists $0 < M \in \mathbb{R}$ such that $|b_n| \leq M$ for all n . Suppose $\sum |a_n|$ converges, then $\sum |a_n|$ satisfies the Cauchy criterion. Let $\epsilon > 0$, then there exists N such that $m \geq n > N$ implies

$$\left| \sum_{k=m}^n |a_k| \right| < \frac{\epsilon}{M}.$$

Consider the sequence $\sum a_n b_n$, we have,

$$\left| \sum_{k=m}^n a_k b_k \right| \leq \sum_{k=m}^n |a_k b_k| = \sum_{k=m}^n |a_k| |b_k| < \sum_{k=m}^n |a_k| M = M \sum_{k=m}^n |a_k| = M \left| \sum_{k=m}^n |a_k| \right| < M \frac{\epsilon}{M} = \epsilon.$$

Therefore $\sum a_n b_n$ satisfies the Cauchy criterion and is therefore convergent. $\square$

Problem 5: (Ross 14.7)

Theorem. *If $\sum a_n$ is a convergent series of nonnegative numbers and $p > 1$, then $\sum a_n^p$ converges.*

Proof. Since $\sum a_n$ converges, $\lim a_n = 0$, by Corollary 14.5, so there exists N such that if $n > N$, $|a_n - 0| = a_n < 1$. Since $p > 1$, We have $a_n^p = a_n a_n^{p-1} < a_n$ for $n > N$. Thus, by the Comparison Test, $\sum_{k=N+1}^{\infty} a_n^p$ converges, and so $\sum a_n^p$ also converges. $\square$

Problem 6: Prove the following theorem using the $\epsilon - \delta$ definition of the limit of a function.

Theorem. *If $\lim_{x \rightarrow a} f(x) = L$, then $\lim_{x \rightarrow a} 4f(x) = 4L$.*

Proof. Let $\epsilon > 0$. Since $\lim_{x \rightarrow a} f(x) = L$, there exists δ such that if $0 < |x - a| < \delta$, then $|f(x) - L| < \epsilon/4$. Assume $0 < |x - a| < \delta$, then

$$|4f(x) - 4L| = |4(f(x) - L)| = 4|f(x) - L| < 4 \cdot \frac{\epsilon}{4} = \epsilon.$$

Since $\epsilon > 0$ was arbitrary, $\lim_{x \rightarrow a} 4f(x) = 4L$. $\square$